FOR YOUR PHYSICIAN

# Dietary and Metabolic Management of ADPKD

## A Summary of Published Research for Clinical Consideration

*Prepared by Santa Barbara Nutrients, Inc. Research conducted at the University of California, Santa Barbara and collaborating institutions. This document is provided for informational purposes and is not a substitute for individualized medical judgment.*

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**I have autosomal dominant polycystic kidney disease (ADPKD), and I'm sharing this summary of published research on dietary and metabolic approaches to the disease because I would like to discuss whether they might be appropriate for me.**

A dietary or metabolic approach may be worth considering across the full range of ADPKD patients. Many people already on tolvaptan combine it with a dietary strategy, and the two can be used together. It's also an option for those not taking tolvaptan—whether due to intolerance, contraindications, lack of insurance coverage, side-effect burden, or personal choice.

Over the past several years, a body of peer-reviewed research has examined carbohydrate restriction and ketogenic metabolic therapy in ADPKD, including a randomized controlled trial in patients. The attached summary lists the key publications so you can review the evidence directly. I'm not asking you to endorse any particular product or protocol—only to consider the published data and whether a dietary or metabolic approach might be suitable in my case.

I understand this is an area of active research rather than established standard of care, and I'd value your assessment.

> **Attached:** a curated list of key publications, with citations and brief summaries. Full publication lists are available at santabarbaranutrients.com/pages/publications and labs.mcdb.ucsb.edu/weimbs/thomas/publications.

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KEY PUBLICATIONS

# Dietary and Metabolic Management of ADPKD

Peer-reviewed research on ketogenic metabolic therapy, ketone and alkaline citrate supplementation, and carbohydrate restriction in ADPKD. Tier 1 is the core clinical evidence; Tier 2 covers the underlying mechanisms and the structured program built on them. Full citations are provided so each can be located on PubMed.

## TIER 1 — CORE CLINICAL EVIDENCE (HUMAN)

### RANDOMIZED CONTROLLED TRIAL

Cukoski S, Lindemann CH, Arjune S, et al. Feasibility and impact of ketogenic dietary interventions in polycystic kidney disease: KETO-ADPKD—a randomized controlled trial. *Cell Reports Medicine*. 2023;4(11):101283. doi.org/10.1016/j.xcrm.2023.101283.

The first randomized controlled trial of ketogenic metabolic therapy in ADPKD (NCT04680780; 66 patients). Kidney function improved significantly, measured by both serum creatinine and cystatin C; a post-hoc analysis found a statistically significant reduction in total kidney volume. The ketogenic diet was feasible for the large majority of participants.

### REAL-WORLD OUTCOMES WITH A MEDICAL FOOD

Muensterman E, Torres J, Weimbs T. Real-world outcomes of managing autosomal-dominant polycystic kidney disease using a medical food as part of a nutrition and lifestyle program to improve renal and metabolic health. *Frontiers in Nutrition*. 2026;12:1624639. doi.org/10.3389/fnut.2025.1624639

Real-world outcome data from ADPKD patients using a medical food (containing exogenous ketones and alkaline citrate) within a structured nutrition and lifestyle program, reporting improvements in renal function, kidney pain, hypertension and metabolic health.

### LONG-TERM KETOGENIC METABOLIC THERAPY WITH EXOGENOUS KETONES AND ALKALINE CITRATE

Messing M, Muensterman E, Marcon S, Weimbs T. Halting cyst progression in ADPKD using long-term ketogenic metabolic therapy and supplementation with exogenous ketones and alkaline citrate—a case series. *Frontiers in Nutrition*. 2026;13:1843178. doi.org/10.3389/fnut.2026.1843178

A case series following ADPKD patients on long-term ketogenic metabolic therapy combined with exogenous ketones and alkaline citrate—the combination underlying the medical-food approach—reporting halted cyst progression and kidney growth.

### RETROSPECTIVE CASE SERIES (FEASIBILITY, SAFETY, AND EFFECTS)

Strubl S, Oehm S, Torres JA, et al. Ketogenic dietary interventions in autosomal dominant polycystic kidney disease—a retrospective case series study: first insights into feasibility, safety and effects. *Clinical Kidney Journal*. 2022;15(6):1079–1092. doi:10.1093/ckj/sfab162.

Patient-reported data from 131 ADPKD patients on self-initiated ketogenic dietary interventions, including improved kidney function and reduced pain, with the diets reported as overall beneficial, safe, and feasible.

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## TIER 2 — MECHANISM AND STRUCTURED PROGRAM

### THE REN-NU DIETARY PROGRAM

Bruen DM, Kingaard JJ, Munits M, Paimanta CS, Torres JA, Saville J, Weimbs T. Ren-Nu, a dietary program for individuals with autosomal-dominant polycystic kidney disease implementing a sustainable, plant-focused, kidney-safe, ketogenic approach with avoidance of renal stressors. *Kidney and Dialysis.* 2022;2:183–203. mdpi.com/2673-8236/2/2/20

The first description of the Ren-Nu program: a sustainable, kidney-safe ketogenic approach designed specifically for ADPKD and structured to avoid known renal stressors.

### MECHANISM — KETOSIS (PRECLINICAL)

Torres JA, Kruger SL, Broderick C, et al. Ketosis ameliorates renal cyst growth in polycystic kidney disease. *Cell Metabolism.* 2019;30(6):1007–1023.e5. doi.org/10.1016/j.cmet.2019.09.012

In mouse, rat, and cat models of PKD, ketosis inhibited disease progression, and adding the ketone β-hydroxybutyrate to a normal diet reproduced the benefit—establishing the metabolic basis of the approach.

### MECHANISM — CRYSTAL DEPOSITION AND CITRATE

Torres JA, Rezaei M, Broderick C, et al. Crystal deposition triggers tubule dilation that accelerates cystogenesis in polycystic kidney disease. *J Clin Invest.* 2019;129(10):4506–4522. doi.org/10.1172/JCI128503

Renal microcrystal deposition was shown to accelerate cyst growth and disease progression; in an ADPKD patient cohort, lower urinary citrate—a natural inhibitor of calcium crystal formation—correlated with greater disease severity. This is the basis for alkaline citrate supplementation as a complementary mechanism alongside ketosis.

### MECHANISM — Β-HYDROXYBUTYRATE + CITRATE COMBINATION (PRECLINICAL)

Torres JA, Holznecht N, Asplund DA, et al. A combination of β-hydroxybutyrate and citrate ameliorates disease progression in a rat model of polycystic kidney disease. *Am J Physiol Renal Physiol.* 2024;326(3):F352–F368. doi.org/10.1152/ajprenal.00205.2023

Preclinical evidence that a defined β-hydroxybutyrate/citrate combination slows disease progression in a PKD rat model, tying the two mechanisms together in a single formulation.

## Review Articles

Messing M, Torres JA, Holznecht N, Weimbs T. Trigger warning: how modern diet, lifestyle, and environment pull the trigger on autosomal dominant polycystic kidney disease progression. *Nutrients.* 2024;16(19). doi.org/10.3390/nu16193281

A review of how modern diet, lifestyle, and environmental factors influence ADPKD progression—useful as a single overview of the framework behind the dietary and metabolic approach.

Torres JA, Sharpe EH, Weimbs T. β-hydroxybutyrate: A Renoprotective Hormone In Polycystic Kidney Disease. *Am J Physiol Endocrinol Metab.* 2026, Jul 30. doi.org/10.1152/ajpendo.00067.2026

A narrative review of the mechanisms underlying the renoprotection by the ketone β-hydroxybutyrate.

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